

CS 5187: Vision and Image
Lecture 9
Stereo Vision

Dr. Dapeng Oliver Wu
City University of Hong Kong
Department of Computer Science

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Amount of horizontal movement is

...

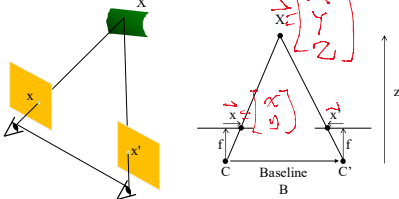
...inversely proportional to the distance from the camera



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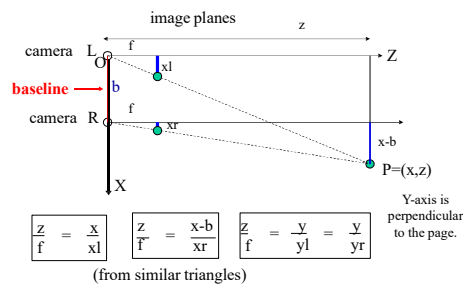
Depth from Stereo

- Goal: recover depth by finding image coordinate x' that corresponds to x



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Optic axes of 2 cameras are parallel



3D from Stereo Images

For stereo cameras with parallel optical axes, focal length f , baseline b , corresponding image points (x_l, y_l) and (x_r, y_r) , the location of the 3D point can be derived from previous slide's equations:

$$\text{Depth } z = f^2 b / (x_l - x_r) = f^2 b / d$$

$$x = x_l z / f \quad \text{or} \quad b + x_r z / f$$

$$y = y_l z / f \quad \text{or} \quad y_r z / f$$

Note that depth is inversely proportional to disparity

Quiz

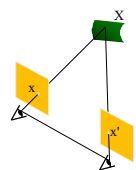
Assume: 1) the optical axes of 2 cameras are parallel;
2) the intrinsic parameters of both cameras are unknown.
If the corresponding positions of scene points in two images are known, which of the following are true?

- Enough to estimate the distance of the scene points to the camera
- Enough to estimate the relative distance (disparity) between scene points in two images
- Enough to estimate the baseline of two cameras

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Depth from Stereo

- Goal: recover depth by finding image coordinate x' that corresponds to x
- Sub-Problems
 - Calibration: How do we recover the relation of the cameras (if not already known)?
 - Correspondence: How do we search for the matching point x' ?



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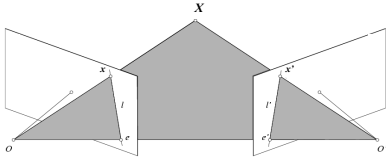
Correspondence Problem



- We have two images taken from cameras with different intrinsic and extrinsic parameters
- How do we match a point in the first image to a point in the second? How can we constrain our search?

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Epipolar constraint: Calibrated case



- Assume that the intrinsic and extrinsic parameters of the cameras are known
- Pixel locations are in normalized image coordinate
- We can set the world coordinate system to the coordinate system of the first camera. Then the projection matrices of the two cameras can be written as $[I | 0]$ and $[R | t]$

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Normalized Image Coordinate

The image coordinate is "normalized" to be independent of camera parameters

$$P' = K[R \quad T]P_w = MP_w$$

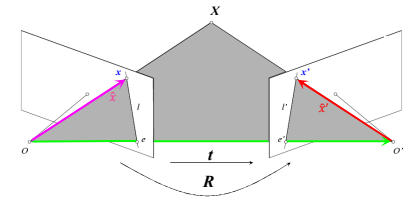
extrinsic parameters
intrinsic parameters

$$K^{-1}P' = [R \quad T]P_w \quad K^{-1}M = [R \quad T] = [I \quad 0] \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}$$

normalized image coordinate projection matrix for normalized camera

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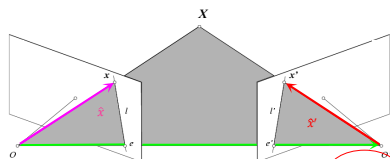
Epipolar constraint: Calibrated case



The vectors $R\hat{x}$, t , and $\hat{x}'$ are coplanar

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Epipolar constraint: Calibrated case



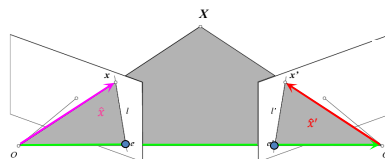
- $E\hat{x}$ is the epipolar line associated with $\hat{x}$ ($l' = E\hat{x}$)
- $Ee = 0$ and $E'e = 0$

Essential Matrix E
(Longuet-Higgins, 1981)

The vectors $R\hat{x}$, t , and $\hat{x}'$ are coplanar

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Epipolar constraint: Uncalibrated case



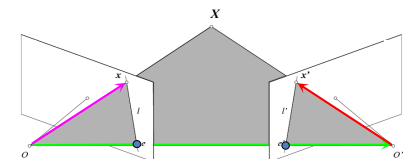
- The calibration matrices K and K' of the two cameras are unknown
- We can write the epipolar constraint in terms of *unknown* normalized coordinates:

$$\hat{x}'^T E \hat{x} = 0 \quad \hat{x}' = K'^{-1}x' \quad \hat{x} = K^{-1}x$$

Un-normalized image coordinate
Normalized image coordinate

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Epipolar constraint: Uncalibrated case



$$\hat{x}'^T E \hat{x} = 0 \Rightarrow x'^T F x = 0 \quad \text{with } F = K'^{-T} E K^{-1}$$

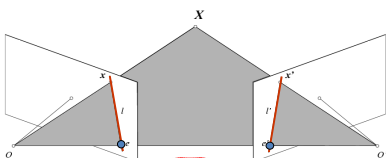
$$\hat{x} = K^{-1}x$$

$$\hat{x}' = K'^{-1}x'$$

Fundamental Matrix
(Faugeras and Luong, 1992)

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Epipolar constraint: Uncalibrated case



$$\hat{x}'^T E \hat{x} = 0 \Rightarrow x'^T F x = 0 \quad \text{with } F = K'^{-T} E K^{-1}$$

- Fx is the epipolar line associated with x ($l' = Fx$)
- F is defined in terms of pixel coordinates

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The eight-point algorithm

$$\begin{aligned} x &= (u, v, 1)^T \\ x' &= (u', v', 1)^T \\ \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} &= 0 \Rightarrow \begin{bmatrix} u'u & u'v & u' & u'v'u & u'v^2 & u'v & v'u & v^2 & u & v & 1 \end{bmatrix} = 0 \\ &A \end{aligned}$$

Minimize:
 $\sum_{i=1}^N (x_i'^T F x_i)^2$
under the constraint $\|F\|^2 = 1$

Smallest eigenvalue of $A^T A$

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Pose Recovery from the Essential Matrix

Essential matrix $E = \hat{T}R$

Theorem 1a (Pose Recovery)

There are two relative poses (R, T) with $T \in \mathbb{R}^3$ and $R \in SO(3)$ corresponding to a non-zero matrix essential matrix.

$$E = U\Sigma V^T$$

$$\begin{aligned} (\hat{T}_1, R_1) &= (UR_{\mathcal{Z}}(+\frac{\pi}{2})\Sigma U^T, UR_{\mathcal{Z}}^T(+\frac{\pi}{2})V^T) \\ (\hat{T}_2, R_2) &= (UR_{\mathcal{Z}}(-\frac{\pi}{2})\Sigma U^T, UR_{\mathcal{Z}}^T(-\frac{\pi}{2})V^T) \end{aligned}$$

$$\Sigma = \text{diag}([1, 1, 0]) \quad R_s(+\frac{\pi}{2}) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Twisted pair ambiguity $(R_2, T_2) = (e^{\hat{G}T}R_1, -T_1)$

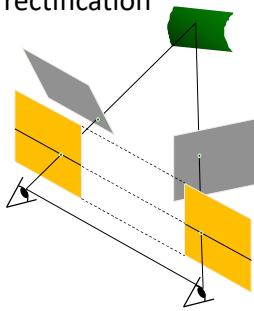
MASKS © 2004

Invitation to 3D vision

Stereo image rectification

- Reproject image planes onto a common plane parallel to the line between camera centers
- Pixel motion is horizontal after this transformation
- Two homographies (3x3 transform), one for each input image reprojection

➤ C. Loop and Z. Zhang, [Computing Rectifying Homographies for Stereo Vision](#), IEEE Conf. Computer Vision and Pattern Recognition, 1999.

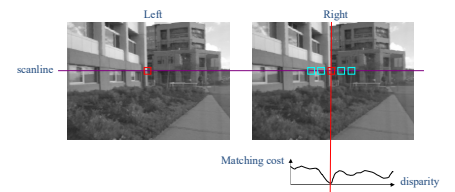


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Example



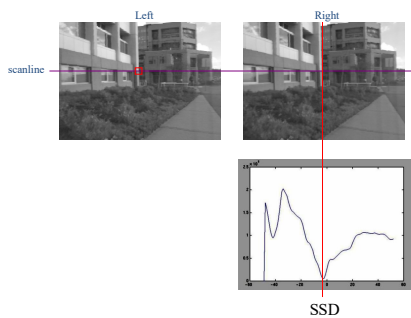
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- Slide a window along the right scanline and compare contents of that window with the reference window in the left image
- Matching cost: SSD, SAD, or normalized cross correlation

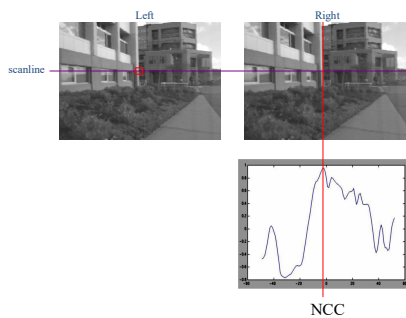
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Correspondence search



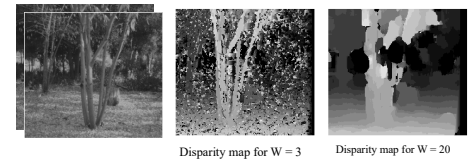
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Correspondence search



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Effect of window size



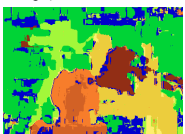
- Smaller window
 - + More details
 - More noise
- Larger window
 - + Smoother disparity maps
 - Less details
 - Fails near boundaries

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Depth map with window-based matching



Depth map (window-based matching) Ground truth for depth map



How can we improve window-based matching?

- So far, matches are independent for each point
- What constraints can we add?

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Constraints

- Uniqueness
 - For any point in one image, there should be at most one matching point in the other image
- Smoothness
 - We expect disparity values to change slowly (for the most part)

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